

CSA 1718 – Artificial Intelligence

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2. Prolog Program to Implement Family Tree

Aim

To write and implement a Prolog program to represent a family tree and identify family relationships.

Objective

- To understand relationships using Prolog.
- To represent parent, father, mother, and grandparent relationships.
- To retrieve family relationships using queries.

Theory

A family tree can be represented using facts such as parent, male, and female. Rules can then be used to derive relationships such as father, mother, and grandparent.

Algorithm

1. Define parent relationships.
2. Define male and female members.
3. Define the rule for father.
4. Define the rule for mother.
5. Define the rule for grandparent.
6. Query the required relationship.

Program

```
parent(john, mary).
```

```
parent(john, david).
```

```
parent(mary, susan).
```

```
parent(david, tom).
```

```
male(john).
```

male(david).

male(tom).

female(mary).

female(susan).

father(X, Y) :-

parent(X, Y),

male(X).

mother(X, Y) :-

parent(X, Y),

female(X).

grandparent(X, Z) :-

parent(X, Y),

parent(Y, Z).

Query

father(john, mary).

Output

true.

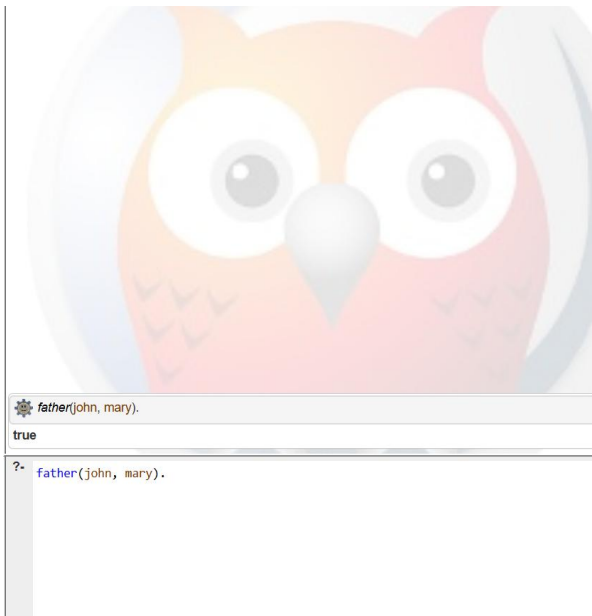
Another query:

grandparent(john, susan).

Output:

true.

```
1 parent(john, mary).
2 parent(john, david).
3 parent(mary, susan).
4 parent(david, tom).
5
6 male(john).
7 male(david).
8 male(tom).
9
10 female(mary).
11 female(susan).
12
13 father(X, Y) :-
14     parent(X, Y),
15     male(X).
16
17 mother(X, Y) :-
18     parent(X, Y),
19     female(X).
20
21 grandparent(X, Z) :-
22     parent(X, Y),
23     parent(Y, Z).
```



Result

Thus, the family tree was successfully represented and relationships were retrieved using Prolog.